

Sunrise Institute of Technology

Syllabus — Introduction to Artificial Intelligence (CS-341)

This is a fictional sample syllabus document created for a student RAG project. It describes a fictional course, CS-341: Introduction to Artificial Intelligence, offered to third-year Computer Engineering students.

Course Overview

CS-341 introduces students to the foundational concepts of artificial intelligence, including search algorithms, knowledge representation, machine learning basics, and an introduction to natural language processing. The course carries 4 credits and is taught over 15 weeks, with 3 lecture hours and 2 laboratory hours per week.

By the end of the course, students should be able to formulate a problem as a search problem, explain the difference between supervised and unsupervised learning, and build a small machine learning model using a standard Python library.

Unit 1: Introduction and Search Algorithms (Weeks 1-3)

Topics covered: history and definitions of artificial intelligence, intelligent agents, problem formulation, uninformed search strategies such as breadth-first search and depth-first search, and informed search strategies such as greedy best-first search and the A* algorithm.

Unit 2: Knowledge Representation and Reasoning (Weeks 4-6)

Topics covered: propositional logic, first-order logic, knowledge bases, forward and backward chaining, and an introduction to rule-based expert systems.

Unit 3: Introduction to Machine Learning (Weeks 7-10)

Topics covered: supervised learning versus unsupervised learning, linear regression, logistic regression, decision trees, an overview of neural networks, and evaluation metrics such as accuracy, precision, and recall.

Unit 4: Natural Language Processing Basics (Weeks 11-13)

Topics covered: text preprocessing, tokenization, word embeddings at a conceptual level, and a brief introduction to how large language models are used for tasks such as question answering.

Unit 5: Applications and Ethics (Weeks 14-15)

Topics covered: real-world applications of artificial intelligence in healthcare, finance, and education, along with a discussion of bias, fairness, and responsible use of AI systems.

Reference Books

1. Artificial Intelligence: A Modern Approach, by Stuart Russell and Peter Norvig, latest edition.
2. Pattern Recognition and Machine Learning, by Christopher Bishop.
3. Speech and Language Processing, by Daniel Jurafsky and James H. Martin.

Laboratory Component

The laboratory component requires students to implement small programs in Python covering each unit, such as implementing the A* search algorithm, building a simple rule-based system, training a basic classifier on a provided dataset, and building a small text classification pipeline.

Laboratory work is evaluated continuously across the semester and contributes to the Laboratory / Practical Work component described in the Course Handbook.

Assessment Summary

Assessment for this course follows the standard institute grading scheme described in the Course Handbook, comprising assignments, a mid-semester test, laboratory work, and an end-semester examination.

Students are advised to refer to the Course Handbook for the exact weight of each component and the minimum passing criteria, since the syllabus document itself does not repeat those details.

Prerequisites

Students registering for CS-341 are expected to have completed CS-210: Data Structures and Algorithms and MA-220: Probability and Statistics in an earlier semester, since concepts from both courses, particularly graph traversal and basic probability, are used throughout the search and machine learning units.

Basic proficiency in Python programming is assumed. Students who have not programmed in Python before are strongly advised to complete a short self-paced Python refresher, links to which are shared on the Learning Portal before the semester begins.

Software and Tools Used

All laboratory work for this course is carried out in Python 3, using standard libraries such as NumPy for numerical computation and scikit-learn for the machine learning unit. Students may use either a local installation or the computer laboratory machines, which have all required libraries pre-installed.

For the natural language processing unit, students will additionally be introduced to basic use of a hosted language model API for a short guided exercise, with clear instructions provided in the laboratory manual regarding what data may and may not be sent to any external service.

Course Coordinator

CS-341 is coordinated by the Department of Computer Engineering. Questions about the syllabus content, unit sequencing, or laboratory exercises for this specific course should be directed to the course coordinator during the published office hours, rather than to the general Academic Office.